

AKHILESH M P

Machine Learning Engineer | AI Engineer | Full-Stack Developer
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PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer and Full-Stack Developer pursuing B.E. in CSE (AI & ML) at Sri Eshwar College of Engineering (2024–2028), with hands-on experience across machine learning, computer vision, LLM-based agentic AI, RAG, and MERN full-stack development. Skilled in Python, TensorFlow/PyTorch, FastAPI, and end-to-end deployment of AI/ML solutions. Seeking an entry-level AI/ML internship to apply strong DSA and applied-ML fundamentals.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, JavaScript, SQL, HTML, CSS

AI / Machine Learning: Deep Learning, Computer Vision, NLP, LLM Agents, RAG, Prompt Engineering, TensorFlow/PyTorch, Scikit-Learn, Keras

Computer Vision: OpenCV, YOLOv8, Object Detection, Image Processing

Generative AI: Gemini API, Groq LLaMA 3.3 70B, Stable Diffusion, ChromaDB

Web & App Development: ReactJS, React Native, Node.js, Express.js, FastAPI, REST APIs, WebSocket, Socket.IO, MERN Stack

Databases: MongoDB, PostgreSQL, ChromaDB, SQL

Data & Visualization: Tableau, Power BI

Tools & Platforms: Git, GitHub, VS Code, Playwright, Raspberry Pi, Canva

Core CS Concepts: Data Structures & Algorithms, OOP, DBMS

PROJECTS

Elephant Detection – Real-Time Wildlife Alert System

2026

Python, YOLOv8, OpenCV, Raspberry Pi, WebSocket, Expo/React Native

- Trained a YOLOv8 computer vision model on a custom Roboflow dataset to detect elephants from a Raspberry Pi camera feed, broadcasting live WebSocket alerts to a mobile app to reduce human-wildlife conflict.
- Deployed the trained model on edge hardware (Raspberry Pi) for on-device inference, avoiding cloud round-trip latency for time-critical alerts.

AEGIS-CPS – Enterprise AI-Powered Cyber-Physical Security Platform

2026

Python, FastAPI, Streamlit, AI Agents, Digital Twin, Machine Learning, Network Security, Incident Response

- Designed and built an enterprise cyber-physical security platform that protects connected robots and industrial systems using a six-layer detection engine, explainable evidence fusion, AI-assisted incident investigation, digital twin monitoring, and live network telemetry.
- Developed an autonomous SOC workflow with AI investigation agents, evidence correlation, human-in-the-loop response approval, and an enterprise command center featuring Fleet Operations, Incident Management, Digital Twin, Live Monitoring, Analytics, and Response Center dashboards.

Agentic AI Social Media Automation

2026

Python, Groq LLM, Stable Diffusion, Playwright

- Developed an autonomous multi-agent system generating captions with Groq LLaMA 3.3 70B and images with a locally-run Stable Diffusion 1.5 pipeline, then auto-publishing across 3 platforms (Facebook, Instagram, LinkedIn) via Playwright.
- Engineered a 3-strategy fallback upload flow and persistent session handling so the agent keeps posting reliably despite frequent platform UI changes.

Nivesh.ai – Multilingual AI Funding Advisor

2025

React, FastAPI, Gemini API, ChromaDB, PostgreSQL

- Built a Gemini-powered Retrieval-Augmented Generation (RAG) platform delivering a 0–100 funding readiness score and a personalised 5-step action checklist for early-stage Indian founders, in their preferred language.
- Designed a ChromaDB vector-retrieval pipeline paired with a PostgreSQL backend to ground funding recommendations in real startup data rather than raw LLM output.

INTERNSHIP EXPERIENCE

Agentic AI Intern – AptitudeGuru

2026

- Studied LLM architecture internals, deep learning concepts (CNN layers, pooling, attention mechanisms), and Transformer fundamentals for building autonomous AI agents.
- Applied theoretical knowledge through hands-on agent-building projects using prompt engineering and multi-agent orchestration.

Stack: Python, LLM Agents, Transformers, Prompt Engineering

MERN Stack Development Intern – EduCentro

2025

- Built and integrated REST APIs for live project modules using MongoDB, Express.js, React, and Node.js.
- Strengthened practical skills in full-stack development, debugging, and production deployment.

Stack: MongoDB, Express.js, React, Node.js, REST APIs

EDUCATION

Sri Eshwar College of Engineering – B.E. CSE (AI & ML) (CGPA: 8.51, up to 4th Semester)

2024 – 2028

ST. ANTONY'S Matric Higher Secondary School – HSC (92.3%)

2023 – 2024

ST. ANTONY'S Matric Higher Secondary School – SSLC (86.6%)

2021 – 2022

CERTIFICATIONS

- Fundamentals of Deep Learning – NVIDIA (2026)
- Data Analytics Virtual Job Simulation – Deloitte (Forage) (2026)
- Design Thinking – NPTEL (2026)
- Introduction to Machine Learning – NPTEL (2025)
- Introduction to Python – Udemy (2025)
- DSA Basics in C++ – Udemy (2025)
- Oracle Badge in Java – Oracle (2025)

ACHIEVEMENTS & CODING PROFILES

- 1st Place – Creatathon (Web Development) | 1st Prize – Project Expo (Design Thinking) | 3rd Place – SCI-FIXX Bug Fixing Competition
- Participated in 4+ national-level hackathons
- LeetCode: 170+ problems solved | SkillRack: 1200+ problems solved